

Markov Chains

Definition

A Markov chain is a stochastic system that transitions from one state to another with specific rule:

- Markov property: system is memory-less. Probability of transitioning depends only on current state.
- Mathematic notation: Sequence of random variables $X_0, X_1, X_2, \dots$ and states $i_0, i_1, \dots$ with:

$$P(X_n = i_n | X_{n-1} = i_{n-1}) = P(X_n = i_n | X_0 = i_0, X_1 = i_1, \dots, X_{n-1} = i_{n-1})$$

Time-homogeneous vs. Time-inhomogeneous

Time-homogeneous (stationary): probability of state transitions is independent of time.

Time-inhomogeneous: probability of moving from one state to another can change with time.

Transition matrices

A transition matrix (P_t) represents the transition probabilities:

$$(P_t)_{i,j} = P(X_{t+1} = j | X_t = i)$$

Multi-step transition by multiplying by transition matrix:

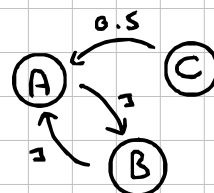
$$(P_{t+k})_{i,j} = P(X_{t+k+1} = j | X_{t+k} = i) \\ = P_t \times P_{t+1} \times \dots \times P_{t+k}$$

Next state		
Current state	0.3 0.5 0.1	$P(A \rightarrow A) = 0.3$
	0.2 0.1 0.6	$P(C \rightarrow B) = 0.4$
	0.6 0.4 0	

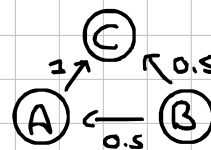
Key properties

Periodicity (k):

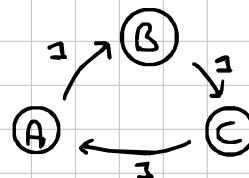
- A state i has a period $k \geq 1$ if any path starting at i and returning to i must take a number of steps divisible by k .
- Aperiodic if $k = 1$, meaning returns can happen at irregular step intervals.



Absorbing state: state i with $(P_t)_{i,i} = 1$.



Irreducibility: if it is possible to transition between two states with a positive probability.



Transience vs. Recurrence:

Transient state: non-zero probability that the system will never return to this state.

Recurrent state: system is guaranteed to return to the state.

- Positive recurrent: expected number of steps is finite.
- Null recurrent: expected number of steps is infinite.

Ergodicity: a state is ergodic if it is both positive recurrent and aperiodic.